Stacking reveals the galaxy baryon fountain

Averaging many simulated galaxies exposes a regular, coherent gas-cycling pattern hidden by individual diversity.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

Authors: Chengyu Ma, Yangyao Chen, Enci Wang, Houjun Mo, Huiyuan Wang, Tao Wang, Qirong Yuan, Min Du et al.

arXiv:2609.21374 doi:10.48550/arXiv.2609.21374

Background

Galaxies grow by cycling gas inward and outward in fountain-like flows. Individual ecosystems are too diverse and dynamic to reveal the underlying pattern directly.

> **Evidence for baryon cycles exists, but no clean, quantitative picture of the average flow geometry has emerged.**

What this paper is selling

The authors stack gas velocity and property fields across galaxy ensembles in a cosmological simulation. This suppresses scatter and exposes strikingly regular fountain patterns matching observations.

Key results to remember

- Stacked fields show smooth, coherent inflow and bipolar outflow cones across the halo.
- A simple parameterized fountain model reproduces the stacked velocity field well.
- The fountain geometry persists across stellar mass and redshift, expanding with halo size.

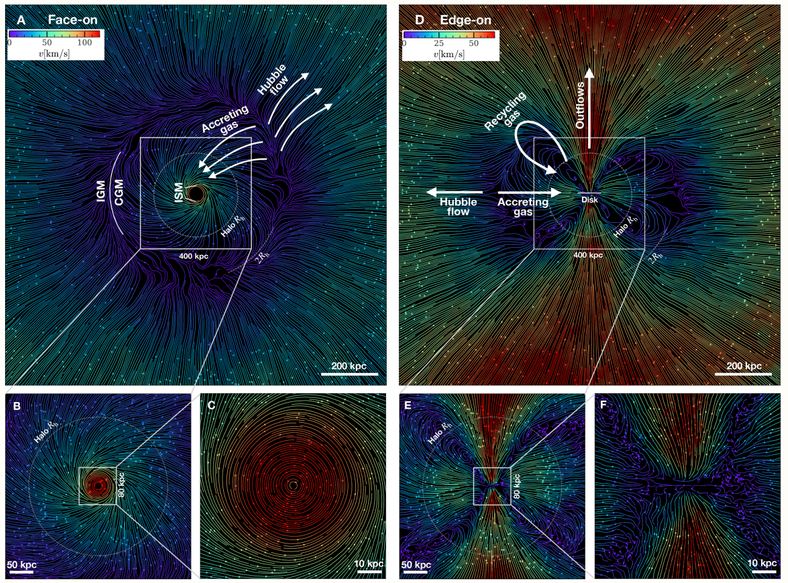


Figure 1 | Stacked streamlines reveal a coherent, fountain-like gas velocity field invisible in any single galaxy.

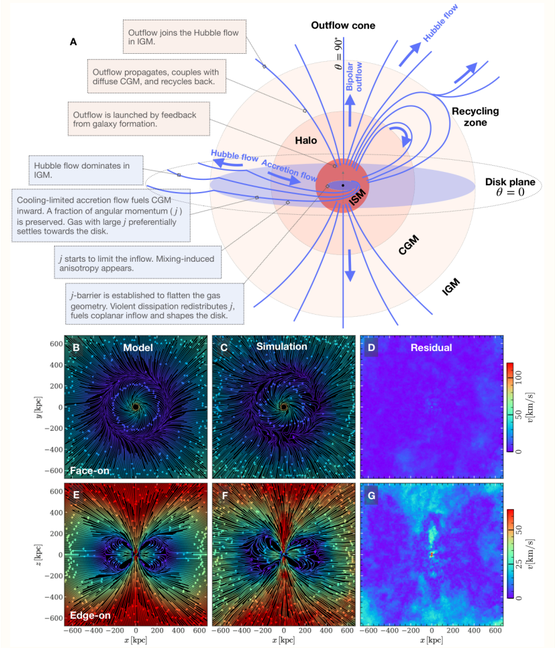


Figure 4 | A compact parameterized fountain model captures the stacked velocity field, with small residuals.